IPhO 1983, Problem T2 (Corrected Solution) — Stefan Ivanov

We will only cover the final two subparts.

(c) My aim here is to show a quick way to obtain a numerical answer for the current $I_{A\to B}$ flowing from node A to node B. I will assume that you're familiar with LC-circuits and the loop current method [1].

Our approach will work only when the numerical values for L_1, L_2, C_1, and C_2 obey $L_1C_1 = L_2C_2$, as is the case in this problem. If we were to cut off the upper half of the circuit from the lower half, both would oscillate with identical angular frequencies $\omega = 1/\sqrt{L_1C_1} = 1/\sqrt{L_2C_2}$. Each half can do its own thing, and the potentials u_A and u_B at nodes A and B will still be able to remain in sync, because both will vary as $\sim \cos(\omega t + \varphi)$. But that's exactly what's supposed to happen in the original circuit, where nodes A and B are equipotential.

This gives us hope that even in the original circuit, the oscillations in the two halves might turn out to be decoupled – in other words, the oscillations at L_1 and C_1 are not influenced at all by what happens with L_2 and C_2. Motivated by this, let's write the current distribution in the circuit as a superposition of three loop currents I_u, I_c, and I_d, as shown on Figure 1. Denote also the charges on the right plates of the capacitors by q_u and q_d, whereby $I_\mathrm{u} = \mathrm{d}q_\mathrm{u}/\mathrm{d}t$ and $I_\mathrm{d} = \mathrm{d}q_\mathrm{d}/\mathrm{d}t$. We'll want to check if there's a valid solution where I_c is constant. After all, that

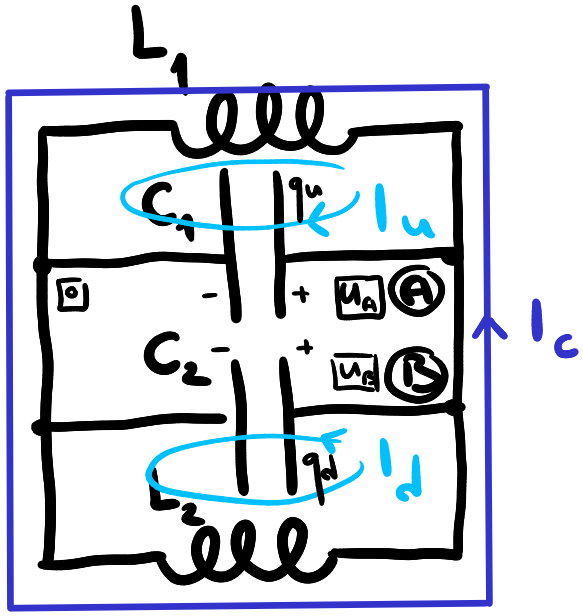


Figure 1. Loop currents.

is what 'decoupled' means, because Kirchoff's voltage law for the upper loop,

$$-L_1\frac{\mathrm{d}(I_\mathrm{u} - I_\mathrm{c})}{\mathrm{d}t} - \frac{q_\mathrm{u}}{C_1} = 0, \tag{1}$$

will reduce to

$$-L_1\frac{\mathrm{d}I_\mathrm{u}}{\mathrm{d}t} - \frac{q_\mathrm{u}}{C_1} = 0 \quad\Longleftrightarrow\quad \ddot{q}_\mathrm{u} + \frac{1}{L_1C_1}q_\mathrm{u} = 0, \tag{2}$$

which indeed depends only on q_u. We'll then have $q_\mathrm{u} = q_{\mathrm{u}0}\cos(\omega t + \varphi_\mathrm{u})$, where the constants q_{u_0} and φ_u depend on the initial conditions. It follows that

$$u_A = \frac{q_\mathrm{u}}{C_1} = \frac{q_{\mathrm{u}0}}{C_1}\cos(\omega t + \varphi_\mathrm{u}). \tag{3}$$

Similarly for the lower loop,

$$u_B = \frac{q_\mathrm{d}}{C_2} = \frac{q_{\mathrm{d}0}}{C_2}\cos(\omega t + \varphi_\mathrm{d}). \tag{4}$$

Our trial solution will be valid so long as $u_A = u_B$ at all times. For this to happen, we want to arrange a value for I_c for which it will turn out that $q_{\mathrm{u}0}/C_1 = q_{\mathrm{d}0}/C_2$ and $\varphi_\mathrm{u} = \varphi_\mathrm{d}$.
From the initial conditions $u_A(0) = u_B(0) = u_0$, we get

$$q_{\mathrm{u}0} \cos\varphi_\mathrm{u} = C_1 u_0, \tag{5}$$
$$q_{\mathrm{d}0} \cos\varphi_\mathrm{d} = C_2 u_0. \tag{6}$$

We also need to match the currents through the inductors L_1 and L_2 to their initial values:

$$I_\mathrm{u}(0) - I_\mathrm{c} = I_{01} \quad \Rightarrow \quad -\omega q_{\mathrm{u}0} \sin\varphi_\mathrm{u} - I_\mathrm{c} = I_{01}, \tag{7}$$
$$I_\mathrm{d}(0) + I_\mathrm{c} = I_{02} \quad \Rightarrow \quad -\omega q_{\mathrm{d}0} \sin\varphi_\mathrm{d} + I_\mathrm{c} = I_{02}. \tag{8}$$

We substitute (5) and (6) into (7) and (8), after which we divide the latter two equations to reach

$$\frac{\tan\varphi_\mathrm{u}}{\tan\varphi_\mathrm{d}} = \left(\frac{C_2}{C_1}\right) \frac{I_{01} + I_\mathrm{c}}{I_{02} - I_\mathrm{c}}. \tag{9}$$

We want an I_c for which the phases are equal. Working in Ampères,

$$1 = \left(\frac{1}{2}\right) \frac{0.1\,\mathrm{A} + I_\mathrm{c}}{0.2\,\mathrm{A} - I_\mathrm{c}} \quad \Rightarrow \quad I_\mathrm{c} = 0.1\,\mathrm{A}. \tag{10}$$

So our guess for a solution ended up being fine. We see that the current flows from node B to node A, so $\boxed{I_{A\to B} = -0.1\,\mathrm{A}.}$

(d) Having obtained I_c, we can also extract the other parameters. Denote $\varphi_\mathrm{d} = \varphi_\mathrm{u} \equiv \varphi$. Using (5) and (7), we have

$$q_{\mathrm{u}0} \cos\varphi = C_1 u_0, \quad q_{\mathrm{u}0} \sin\varphi = -\frac{I_{01} + I_\mathrm{c}}{\omega} \quad \Rightarrow \quad q_{\mathrm{u}0} = \sqrt{(C_1 u_0)^2 + \left(\frac{I_{01} + I_\mathrm{c}}{\omega}\right)^2} = 2.04\,\mu\mathrm{C}. \tag{11}$$

Then, $q_{\mathrm{d}0} = (C_2/C_1) q_{\mathrm{u}0} = 1.02\,\mathrm{mC}$. The phase φ follows from (7):

$$\varphi = -\arcsin\left(\frac{I_{01} + I_\mathrm{c}}{\omega q_{\mathrm{u}0}}\right) = -78.7^\circ. \tag{12}$$

Finally, recall that the current through the inductor L_1 is

$$I_1 = -\omega q_{\mathrm{u}0} \sin(\omega t + \varphi_\mathrm{u}) - I_\mathrm{c}. \tag{13}$$

After plugging in the numbers,

$$I_1 = -(0.204\,\mathrm{A}) \cdot \sin\left((10^5\,\mathrm{rad/s}) \cdot t - 1.37\,\mathrm{rad}\right) - (0.100\,\mathrm{A}). \tag{14}$$

The amplitude of the oscillating term is $\boxed{I_{1\mathrm{A}} = 0.204\,\mathrm{A}.}$

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References

[1] J. Kalda, Electrical Circuits, 2017, p. 7,
https://www.ioc.ee/~kalda/ipho/electricity-circuits.pdf
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